Sensitivity analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Vary the setting. Keep every result. State what is unknown.

Explain sensitivity patterns without turning prediction counts into accuracy or tuning into fresh evaluation.

Keep one habit: Declared range → retrieval tradeoff ledger → fixed-score native grid → equality/selection probes → exact audit → independent pass-count range.

MAIN LESSON · SECTION 1

Change a plausible setting and preserve the complete pattern

> Sensitivity analysis reveals dependence on choices; it does not automatically choose the best setting.

Learning goals

I can explain the **mechanism** and terminology of **sensitivity analysis**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve controls, ablation and evidence limits

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling variability. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting response

- Recall Ablation studies. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

My starting explanation

End goal

Explain **sensitivity analysis** using the records and boundaries of this fictional investigation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **sensitivity analysis**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Separate changing predictions from checking correctness

> Retrieval depth and classification score cutoff are different controls.

Begin with dependence on a choice

A conclusion may depend on a cutoff, retrieval depth or analysis choice. Sensitivity analysis deliberately varies plausible choices while preserving the rest of the comparison. Keep every tested setting and its result, including unfavorable outcomes. The question is how the pattern changes across the declared range.

If you then choose a setting using those outcomes, that becomes tuning; report it and use separately reserved evaluation for a fresh performance claim.

Original sensitivity definition

Sensitivity analysis checks whether a conclusion changes under plausible settings or analysis choices. It should not secretly select the most favourable result.

Full inputs, control types and missing evidence

Sensitivity item	Complete exercise contract
Main retrieval settings	k = 1, 3, 5; number of retrieved items, not a score threshold

Sensitivity item	Complete exercise contract
Supplied recall summaries	0.4, 0.7, 0.8 at the respective settings
Supplied precision summaries	0.9, 0.7, 0.5 at the respective settings
Missing main information	Raw relevance labels, denominators and averaging convention; do not reconstruct counts
Separate score picture	Fixed scores 0.4, 0.6, 0.8; classify positive when score ≥ cutoff
Picture cutoffs	0.5, 0.7 and boundary check 0.6
Missing picture information	No reference labels; accuracy and best cutoff cannot be determined
Interpretation limit	Only supplied settings are described; no beyond-range or universal guarantee

Given and missing facts

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Trace the complete retrieval tradeoff

> A larger retrieval depth improves one supplied summary while reducing another.

Prespecified range and complete-results procedure

Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.

All three settings and observed summary pattern

Retrieved items k	Supplied recall	Supplied precision	Pattern in the tested range
1	0.4	0.9	Lowest recall; highest precision
3	0.7	0.7	Intermediate summaries
5	0.8	0.5	Highest recall; lowest precision
From k = 1 to k = 5	Increase 0.4, or 40 percentage points	Decrease 0.4, or 40 percentage points	Tradeoff; no universal best setting
What remains unknown	Raw labels/counts	Averaging convention	No invented per-query count or significance

State scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Sensitivity analysis — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5.	Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.	Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict the equality boundary without changing scores

> At least includes the score exactly equal to the cutoff.

Main task and response

- Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict picture cutoff 0.6

- Change: Set the cutoff to exactly 0.6.

▼ **Check after predicting**

Prediction: 0.6 and 0.8 are positive because the rule includes equality.

Reason: The declared at-least boundary includes the threshold value.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read each fixed score against every declared cutoff

> Changing a decision boundary is not improving a measured accuracy.

Main retrieval summaries and checked pattern

Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5.

Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate unlabelled-score inputs and scope

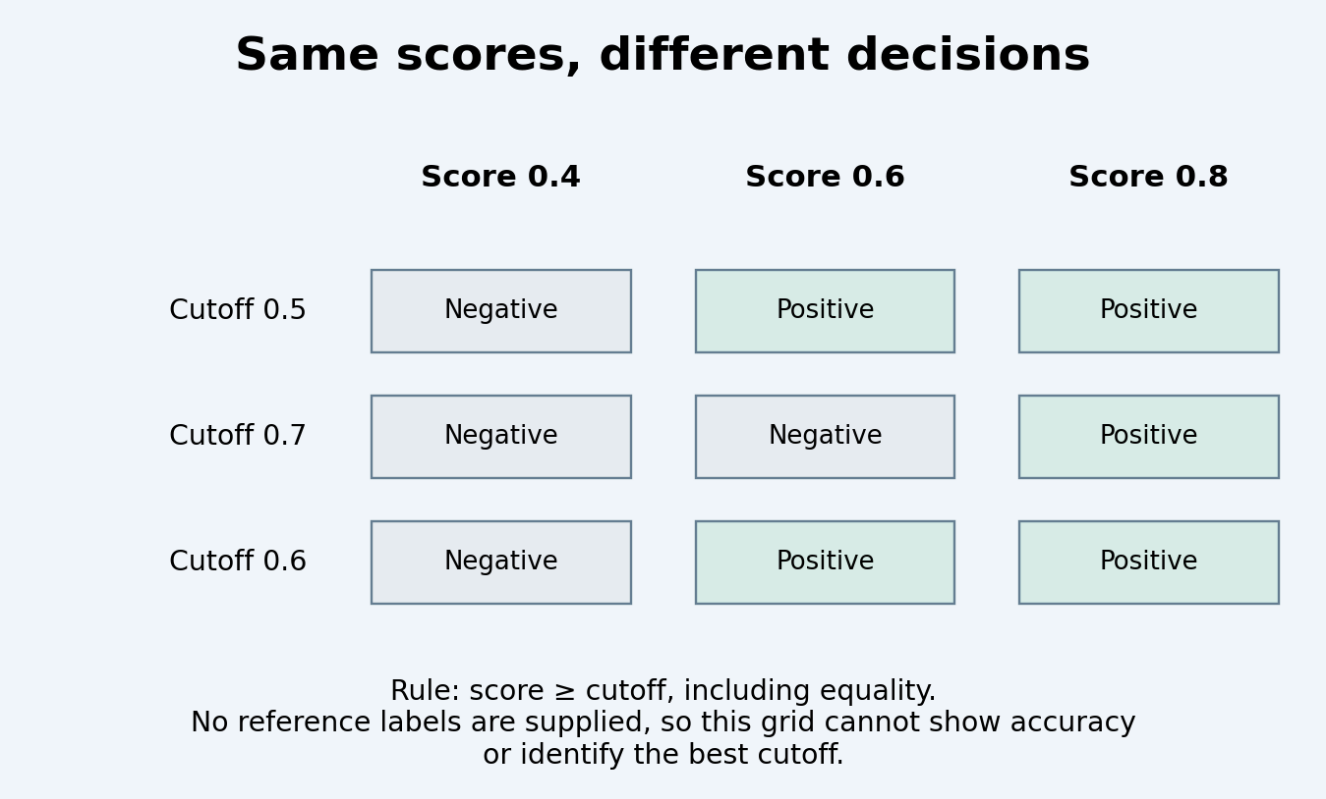
A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

Scores are 0.4,0.6,0.8. Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

The picture fixes three unlabelled scores at 0.4, 0.6 and 0.8. The rule is positive when score is at least the cutoff. At 0.5 there are two positives; at 0.7 there is one; at exactly 0.6 there are two because equality counts. These are prediction flags, not correctness judgments.

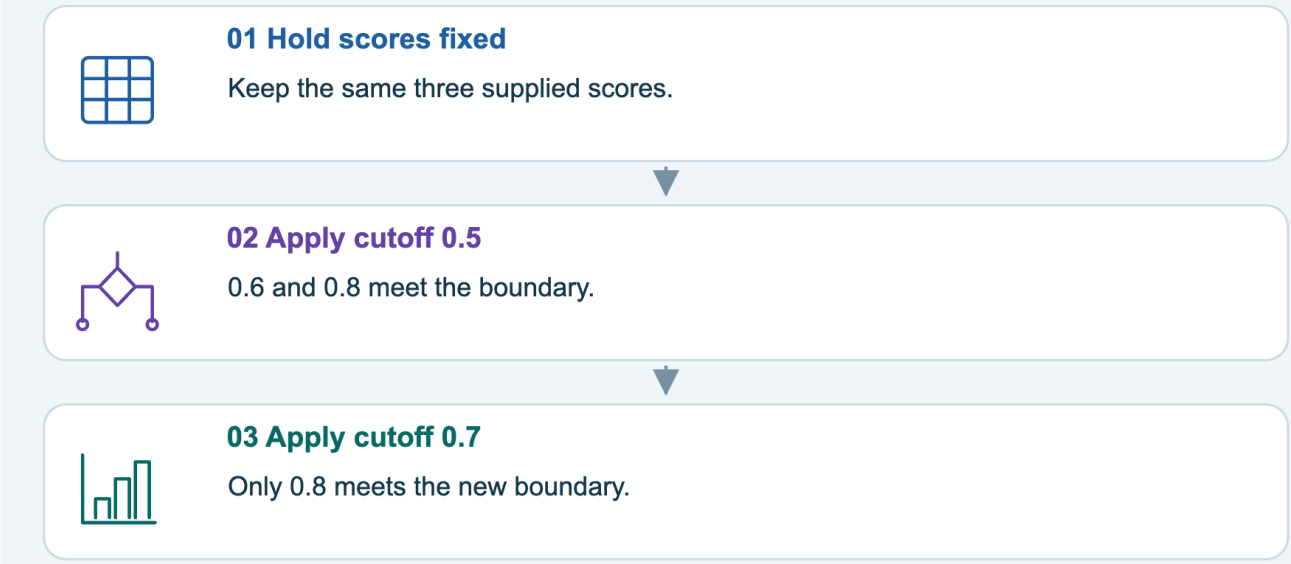
If the unknown labels were negative/positive/positive, the first rule would fit all three; if instead negative/negative/positive, the second would. Those are explicitly hypothetical alternatives, not hidden reference labels. Without actual labels or a task objective, no best cutoff follows.

Inspect the equality row in the native grid



AI ACADEMY / AI-7/17 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Hold scores fixed	Keep the same three supplied scores.
2. Apply cutoff 0.5	0.6 and 0.8 meet the boundary.
3. Apply cutoff 0.7	Only 0.8 meets the new boundary.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

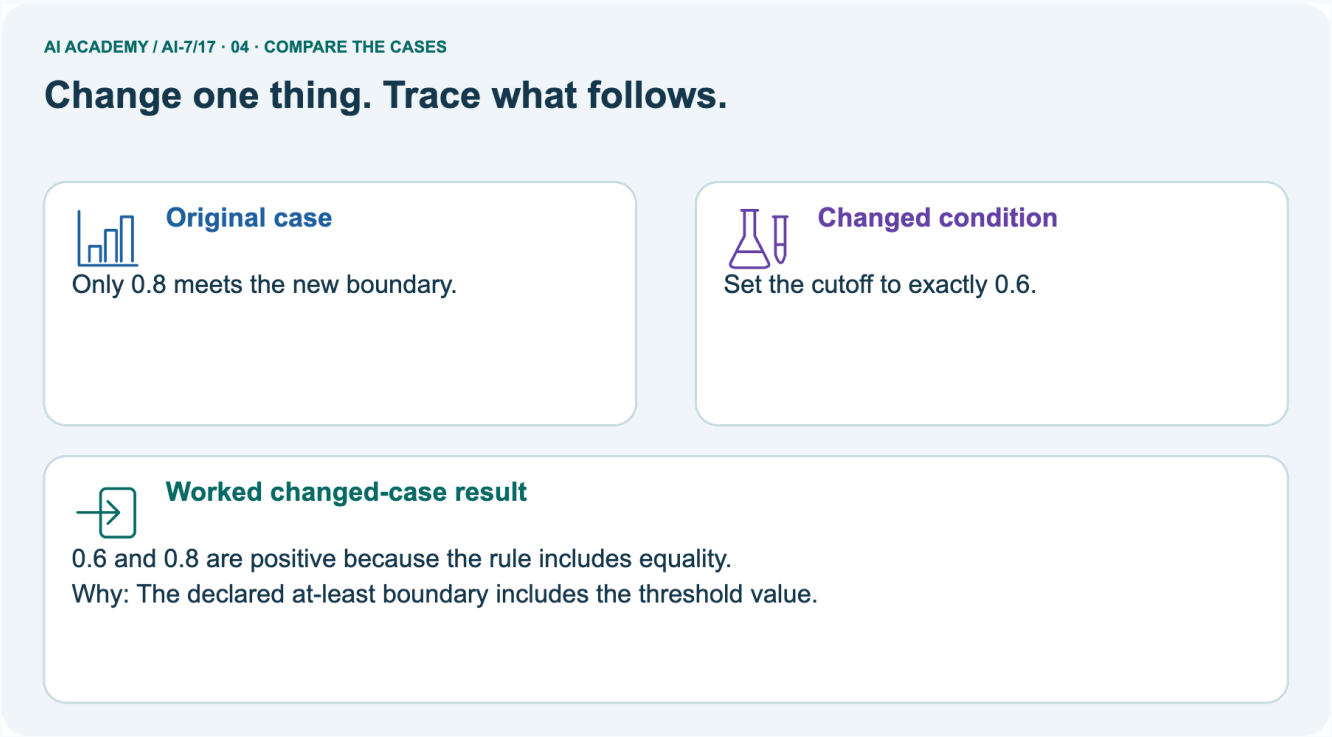
Complete prediction flags and unknown correctness

Fixed score	Cutoff 0.5	Cutoff 0.7	Cutoff 0.6
0.4	Negative	Negative	Negative
0.6	Positive	Negative	Positive: equality included
0.8	Positive	Positive	Positive
Positive count	2 of 3	1 of 3	2 of 3
Correct count	Unknown	Unknown	Unknown
Scores themselves	Unchanged	Unchanged	Unchanged

Predict exact cutoff 0.6

- Change one thing: Set the cutoff to exactly 0.6. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain label and selection limits

- Without reference labels, these counts do not identify the best cutoff or establish correctness.

- Read the labelled picture: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sensitivity analysis

The facts we will use

AI ACADEMY / AI-7/17

Score decisions under inclusive cutoffs

	Score 0.4	Score 0.6	Score 0.8
Cutoff 0.5	No	Yes	Yes
Cutoff 0.7	No	No	Yes
Cutoff 0.6	No	Yes	Yes

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/17

Sensitivity analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Hold scores fixed

Keep the same three supplied scores.



2. Apply cutoff 0.5

0.6 and 0.8 meet the boundary.



3. Apply cutoff 0.7

Only 0.8 meets the new boundary.

CHANGE ONE THING

Set the cutoff to exactly 0.6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair selective reporting and prediction-versus-accuracy confusion

> A sensitivity pattern is conditional on tested choices and valid evaluation roles.

Original limitation and repair setup

A stable tested range does not establish stability under every parameter or dataset change.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete equality, selection and scope probes

Sensitivity audit	Checked interpretation
Report only the best tested setting	Hides sensitivity and outcome-driven selection
More positive predictions	Does not imply more correct predictions
Score equals cutoff	Positive under the stated inclusive rule
Raw scores unchanged	Decisions can still change when the rule changes
Recall increases as precision falls	Describe tradeoff; objective and constraints are needed for selection
Aggregate precision/recall supplied	Do not invent per-query counts or a missing averaging method
A pattern holds at three settings	Does not prove all intermediate or untested settings behave identically
Choose using these results	Disclose tuning and preserve separately reserved evaluation

Sensitivity audit	Checked interpretation
Change data roles or split during sensitivity analysis	Record the change; leakage can invalidate comparison

Write a defensible repair

- A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise keeping all settings and all results visible

> Reserve the independent pass-count example for the first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported workbook trace

Use workbook W2 for the connected example "**Sensitivity analysis**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual check

- Use the worked case for **Sensitivity analysis**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6. Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

My explanation using a pictured step and the given facts

Reserve the selected threshold/pass case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete fixed-score sensitivity investigation

> Use hypothetical counterexamples to explain why missing labels matter.

Current investigation

Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5. Show the intermediate steps and give the conclusion with its scope.

Complete exact-boundary audit

```
"""Sensitivity arithmetic on supplied summaries and unlabelled toy scores."""
from fractions import Fraction

def exact(value):
 if type(value) not in (str,int,Fraction):raise ValueError('use exact strings or
fractions')
 result=Fraction(value)
 if not 0<=result<=1:raise ValueError('value must be in [0,1]')
 return result

def classify(scores,cutoff):
 if not isinstance(scores,(list,tuple)) or not scores:raise ValueError('scores required')
 boundary=exact(cutoff)
 return [exact(score)>=boundary for score in scores]
retrieval=[{'k':1,'recall':Fraction(4,10),'precision':Fraction(9,10)},
           {'k':3,'recall':Fraction(7,10),'precision':Fraction(7,10)},
           {'k':5,'recall':Fraction(8,10),'precision':Fraction(5,10)}]
scores=['.4','.6','.8'];before=scores[:]
picture={cutoff:classify(scores,cutoff) for cutoff in ['.5','.7','.6']}
assert picture['.5']==picture['.6']==[False,True,True]
assert picture['.7']==[False,False,True] and scores==before
# Labels are explicitly hypothetical, not supplied ground truth for the picture.
hypothetical_labels_A=[False,True,True]
hypothetical_labels_B=[False,False,True]
def correct_count(predictions,labels):
 if len(predictions)!=len(labels):raise ValueError('same cases required')
 return sum(a==b for a,b in zip(predictions,labels))
assert correct_count(picture['.5'],hypothetical_labels_A)==3
assert correct_count(picture['.7'],hypothetical_labels_B)==3
print('Supplied retrieval summary tradeoff:',retrieval)
print('Picture positive flags at each cutoff:',picture)
print('No reference labels are supplied; different explicitly hypothetical labels favor
```

```
different cutoffs. More positives is not greater accuracy.')
print('Retrieval precision/recall are stipulated aggregate summaries; averaging convention
and raw counts are not supplied. No raw relevance labels are reconstructed.')
print('Preserve the tested range and all results. Selection on these results needs
separately reserved evaluation; no actual model run or beyond-range guarantee is
supplied.')
```

Predictions and evidence limits

1. Predict the positive flags at all three picture cutoffs, including equality. The complete code retains the supplied main retrieval summaries without inventing underlying labels or counts. It compares fixed unlabelled scores with exact rational boundaries and demonstrates two hypothetical label sets that favor different cutoffs. Neither label set is an actual observation.

   The arithmetic does not run a model, estimate significance or choose a universally best parameter. Keep the independent ten-case pass-count example for a first attempt.

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: summarize the pass-count range across three thresholds

> Preserve the full pattern and state a selection limit.

Independent case

Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Thresholds .4, .5, .6 give passes 8, 9, 7 out of 10. Report range and a selection limit.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the planned range and every tested result

> Disclose selection and retain evaluation roles.

Project and artifact

Add a labelled **sensitivity analysis evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sensitivity analysis**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local trace or calculation

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain sensitivity without claiming an unmeasured best setting

> State which data, settings and conclusions remain fixed or change.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **sensitivity analysis**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Sensitivity analysis	Checking outcomes across declared plausible choices
Robustness range	The conditions over which a conclusion is examined
Post-hoc selection	Choosing a procedure after seeing its scores

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A stable tested range does not establish stability under every parameter or dataset change. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Sensitivity analysis — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Retrieval cutoffs 1, 3, 5 yield recall .4, .7, .8 and precision .9, .7, .5.	Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.	Higher cutoff increases recall here while precision decreases. No universally best cutoff follows without a task-specific criterion.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.

By the end, I can	How I will show it
Explain the mechanism and terminology of sensitivity analysis	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff.
- Which labelled step matches this detail? Only 0.8 meets the new boundary.
- 1. Hold scores fixed
- 2. Apply cutoff 0.7
- 3. Apply cutoff 0.5
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply cutoff 0.7 Only 0.8 meets the new boundary. **Sensitivity analysis** checks how conclusions change under plausible choices.

Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Changed-case question and optional hint

- Changed-case practice: Set the cutoff to exactly 0.6. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. 0.6 and 0.8 are positive because the rule includes equality.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: 0.6 and 0.8 are positive because the rule includes equality. The declared at-least boundary includes the threshold value.

Repair and retain the record

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Statistical tests and assumptions (ai-7/14). Goal: Explain the **mechanism** and terminology of statistical tests and assumptions. Short practice: Calculate each paired difference and the observed sum.

Count null arrangements with absolute sum at least four, compute p, and explain the decision at threshold 0.05.. Return to this week and explain what you changed.

Review Sampling variability (ai-7/11). Goal: Explain the **mechanism** and terminology of sampling variability. Short practice: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

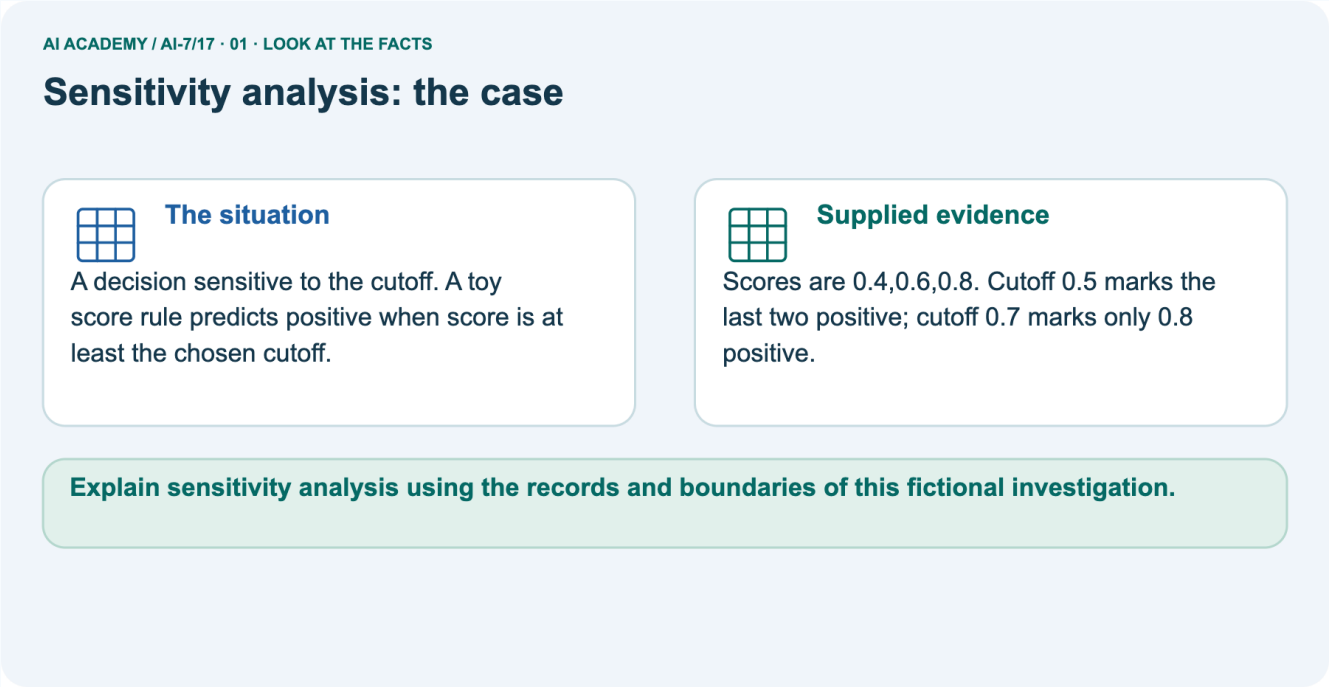
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Sensitivity analysis: the case



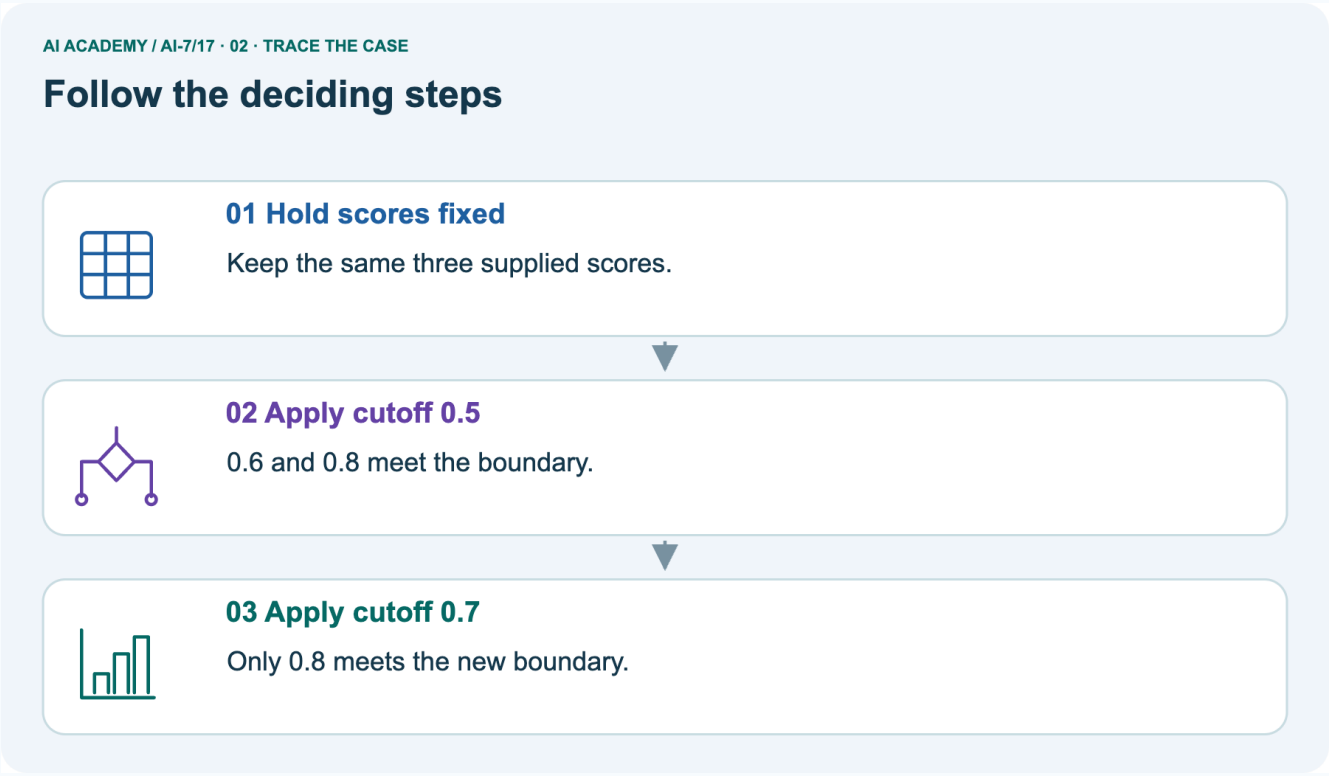
Follow the labelled picture with the matching explanation. Open the vector diagram

A decision sensitive to the cutoff. A toy score rule predicts positive when score is at least the chosen cutoff. Supplied facts: Scores are 0.4,0.6,0.8.

Cutoff 0.5 marks the last two positive; cutoff 0.7 marks only 0.8 positive.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

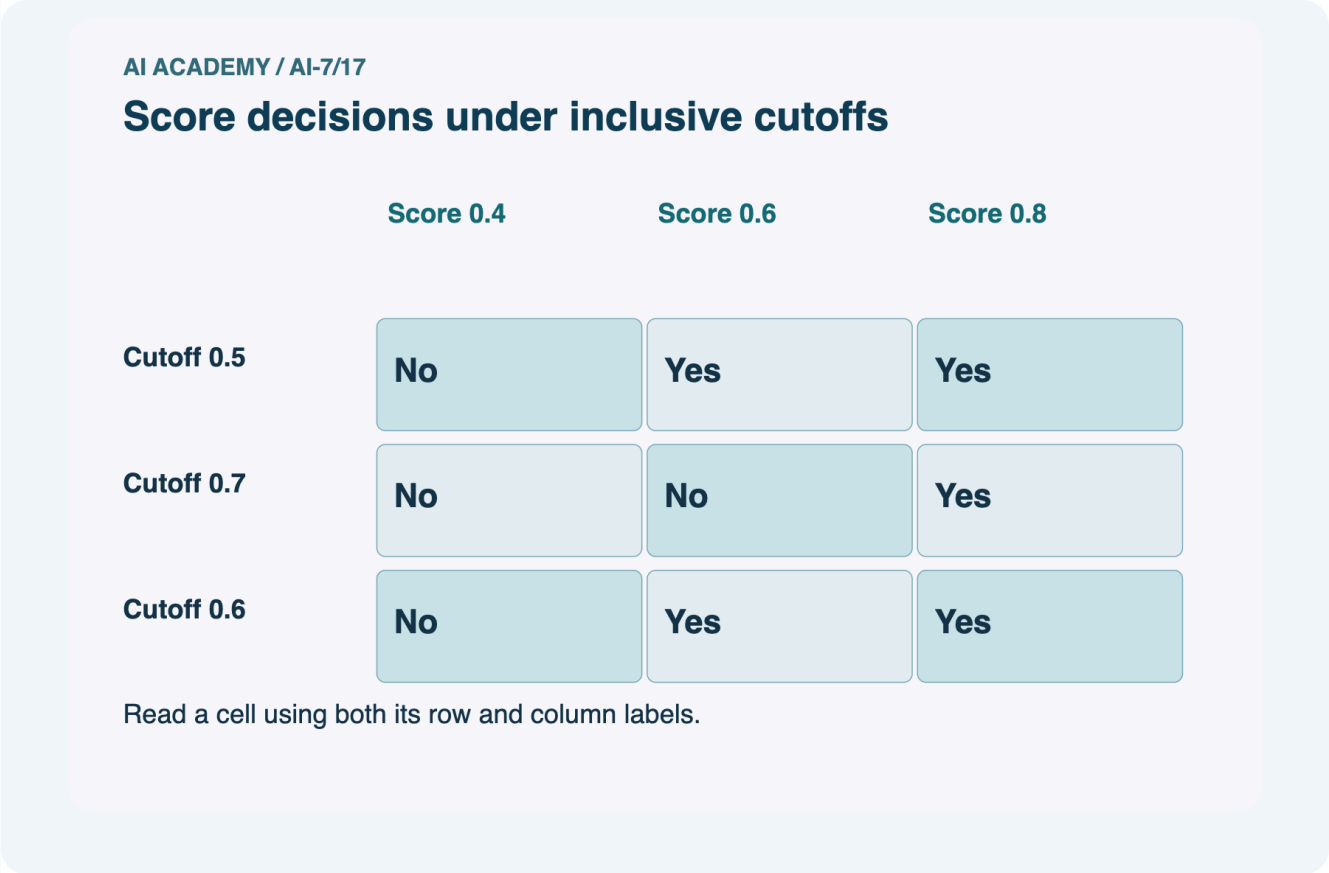


Follow the labelled picture with the matching explanation. Open the vector diagram

Hold scores fixed: Keep the same three supplied scores. → Apply cutoff 0.5: 0.6 and 0.8 meet the boundary. → Apply cutoff 0.7: Only 0.8 meets the new boundary.

Point to each numbered stage and explain which recorded **input** it uses.

03. Score decisions under inclusive cutoffs

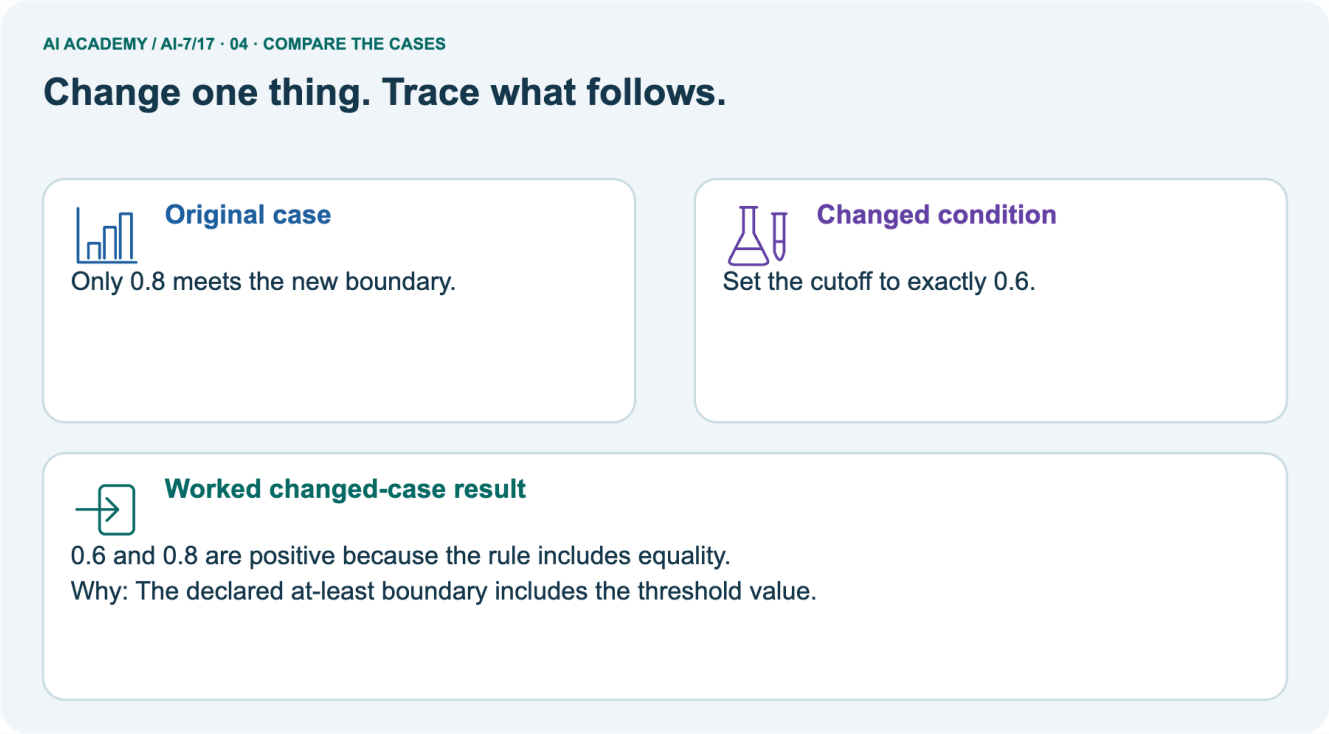


Follow the labelled picture with the matching explanation. Open the vector diagram

Score decisions under inclusive cutoffs. Cutoff 0.5: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes. Cutoff 0.7: Score 0.4 = No, Score 0.6 = No, Score 0.8 = Yes. Cutoff 0.6: Score 0.4 = No, Score 0.6 = Yes, Score 0.8 = Yes.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



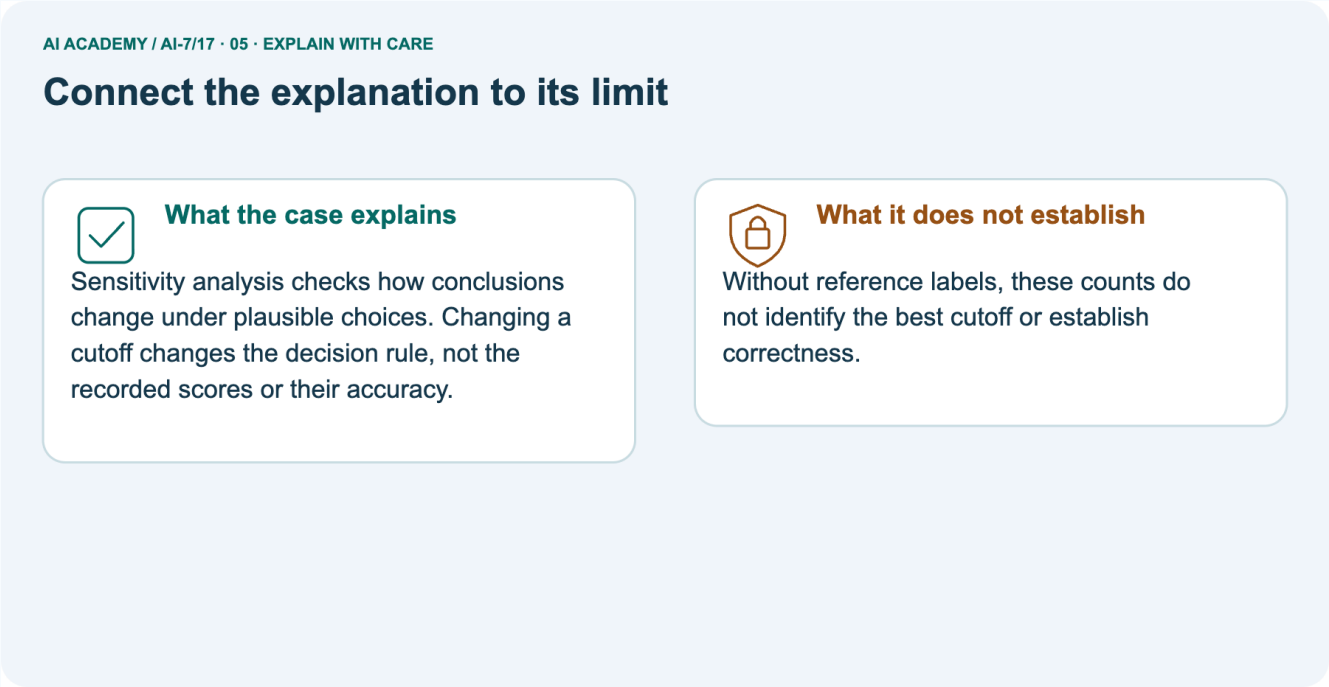
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only 0.8 meets the new boundary. Changed condition: Set the cutoff to exactly 0.6. Expected result: 0.6 and 0.8 are positive because the rule includes equality.

Reason: The declared at-least boundary includes the threshold value.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



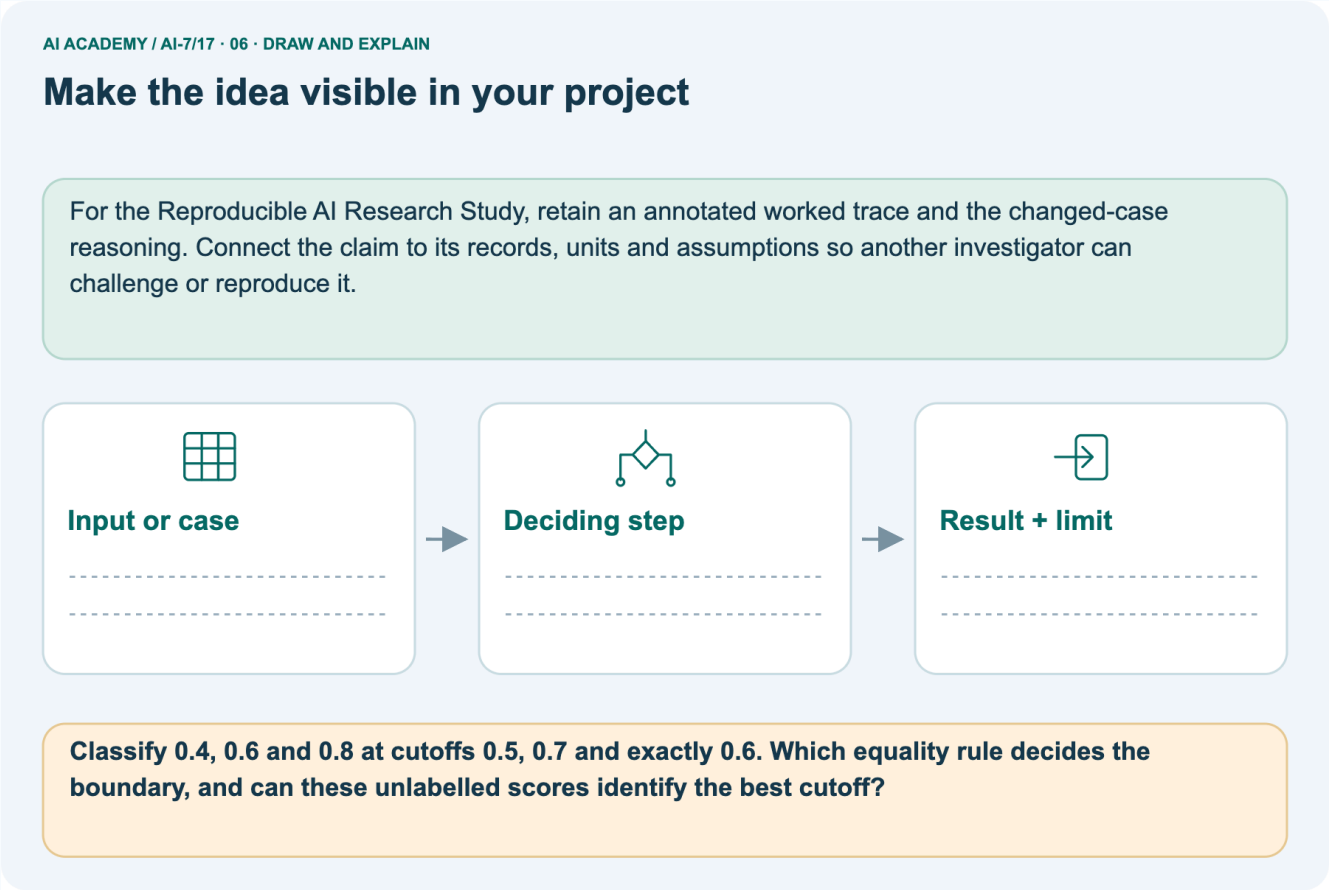
Follow the labelled picture with the matching explanation. Open the vector diagram

Sensitivity analysis checks how conclusions change under plausible choices. Changing a cutoff changes the decision rule, not the recorded scores or their accuracy.

Limit: Without reference labels, these counts do not identify the best cutoff or establish correctness.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Sensitivity analysis: the case

02. Follow the deciding steps

03. Score decisions under inclusive cutoffs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.